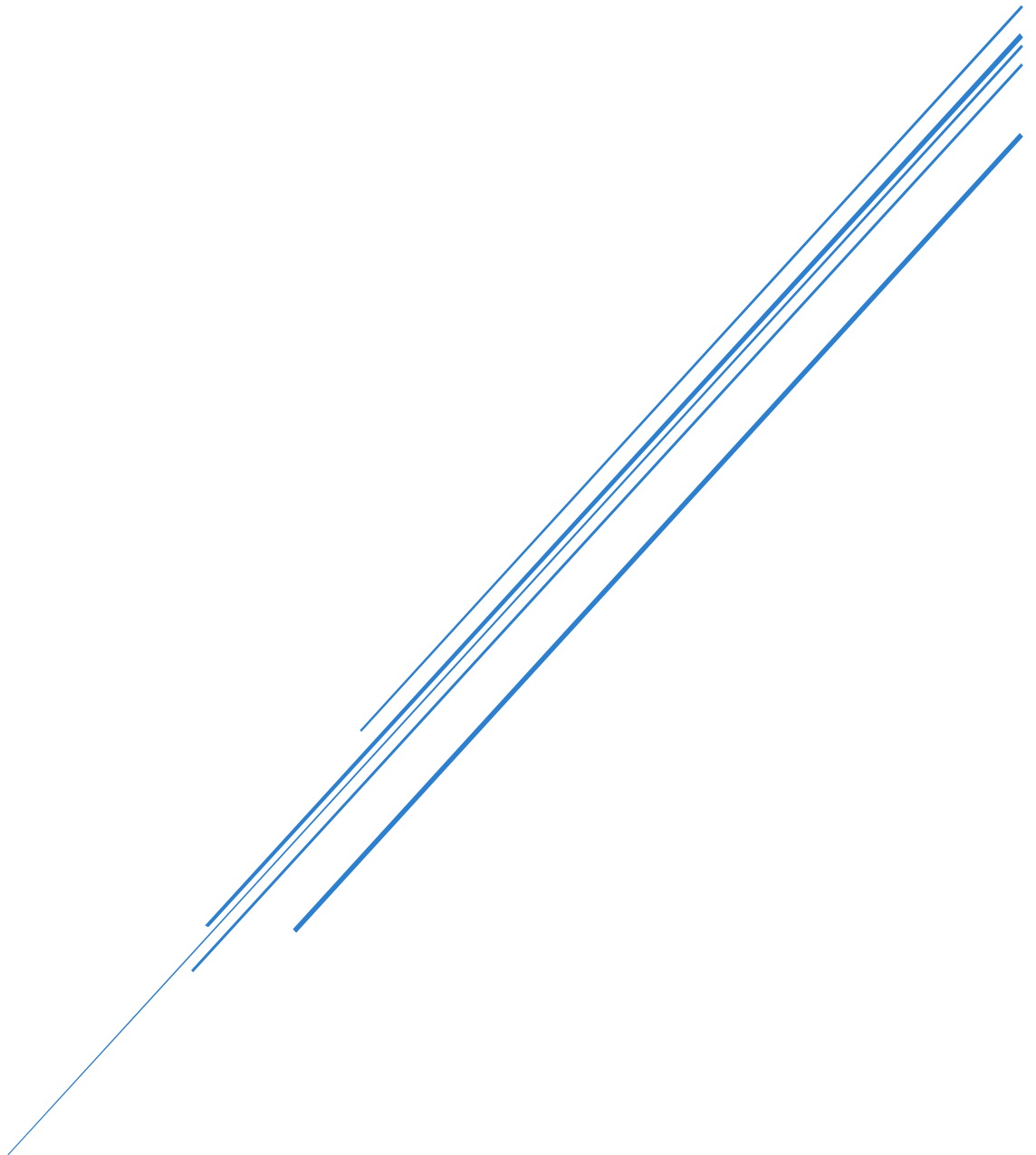


LAB 5 ANSWERS REPORT

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Question 1

Calculate the accuracy of a Logistic Regression model on the Iris dataset that compares the Virginica and Versicolour species with the Setosa species. Report True Positives, True Negatives, False Positives and False Negatives from the confusion matrix.

As seen in the confusion matrix in Figure 1: Confusion Matrix of Iris Dataset we have no false positives or false negatives we perfectly predict all true positives and true negatives 1.0 accuracy. This is likely to the Setosa being cleanly separable to the Virginica and Versicolour species (we've seen this before for the Iris dataset due to certain features being very distinct to the Setosa class).

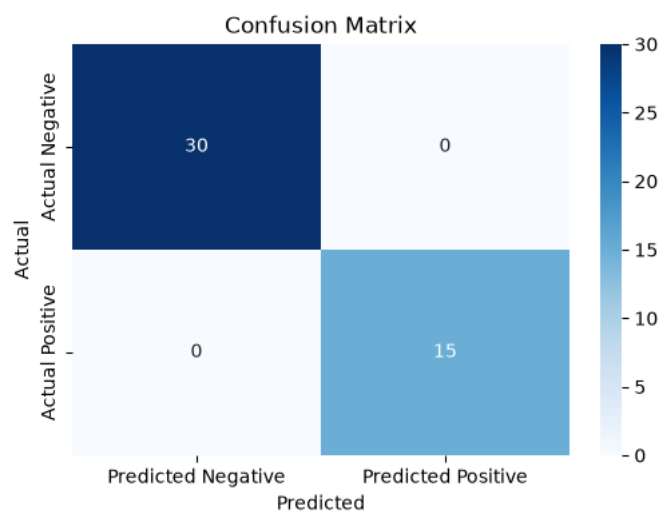


Figure 1: Confusion Matrix of Iris Dataset

Question 2

Calculate and interpret the Precision and Recall across the three classes for a Decision Tree Classifier on the Iris dataset.

As seen in Table 1: Precision and Recall on Iris Dataset the Setosa has perfect precision and recall (not surprising considering the 1.0 accuracy from question 1). Versicolor having 1.0 precision but 0.93 recall shows it had no false positives but did have some false negatives. The contrary is to be said for Virginica since the precision and recall scores basically are mirrored.

Class	Precision	Recall
Setosa	1.0	1.0
Versicolor	1.0	0.93
Virginica	0.94	1.0

Table 1: Precision and Recall on Iris Dataset

Question 3

Plot the ROC Curve and calculate the AUC for a Decision Tree Classifier with a maximum depth of 3, a minimum of 4 samples required to split an internal node, and a minimum of 2 samples required to be at a leaf node on the Iris dataset.

In Figure 2: ROC Curve and AUC Score we can see a perfect ROC curve and AUC score for our decision tree showing no false positives were made. It may be a sign of overfitting (which decision trees are prone to) it's also the Iris dataset so not many instances and a depth of 3 is actually not too shallow for such low dimensional feature space.

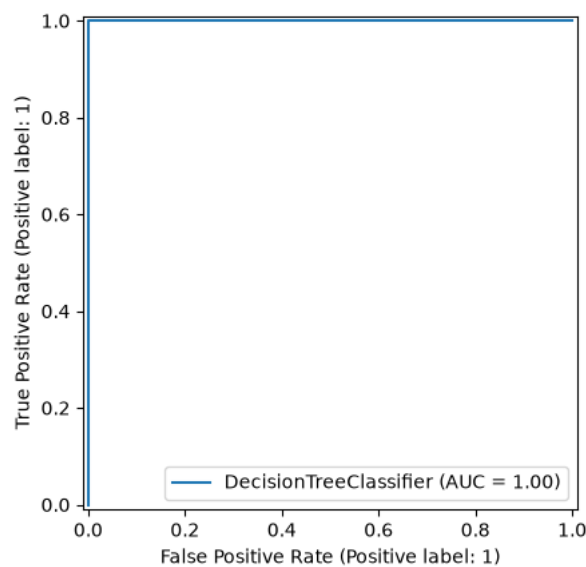


Figure 2: ROC Curve and AUC Score

Question 4

Calculate the Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE) for a Decision Tree Regressor with a maximum depth of 5, a minimum of 10 samples required to split an internal node, and a minimum of 5 samples required to be at a leaf node on the California Housing dataset.

Mean Squared Error: 0.5211334618725708

Mean Absolute Error: 0.5276171441344896

Root Mean Squared Error: 0.7218957416916731

Question 5

Compare the performance of a fully fitted Decision Tree Regressor and an optimally pruned Decision Tree Regressor on the California Housing dataset using R-squared and RMSE. Evaluate the models on both the training data and a hold-out test set. Discuss which model performs better and why.

Test Set Metrics

Pruned R^2 Score: 0.6731656127147243

Pruned Root Mean Squared Error: 0.653168360579731

Full Fitted R^2 Score: 0.5879362320670309

Full Fitted Root Mean Squared Error: 0.7334043511435282

Validation (Hold-out) Set Metrics

Pruned R^2 Score: 0.6907913891669553

Pruned Root Mean Squared Error: 0.6511250927138597

Full Fitted R^2 Score: 0.589235490352986

Full Fitted Root Mean Squared Error: 0.7504730465572462

Discussion

We can see that across both the test and hold-out validation sets that our pruned tree performs slightly better showing that it is probably generalising better as well. But we also see that across the validation and test set there's little variance in the metrics as well showing that the models aren't getting lucky and these scores are a solid evaluation of the models.